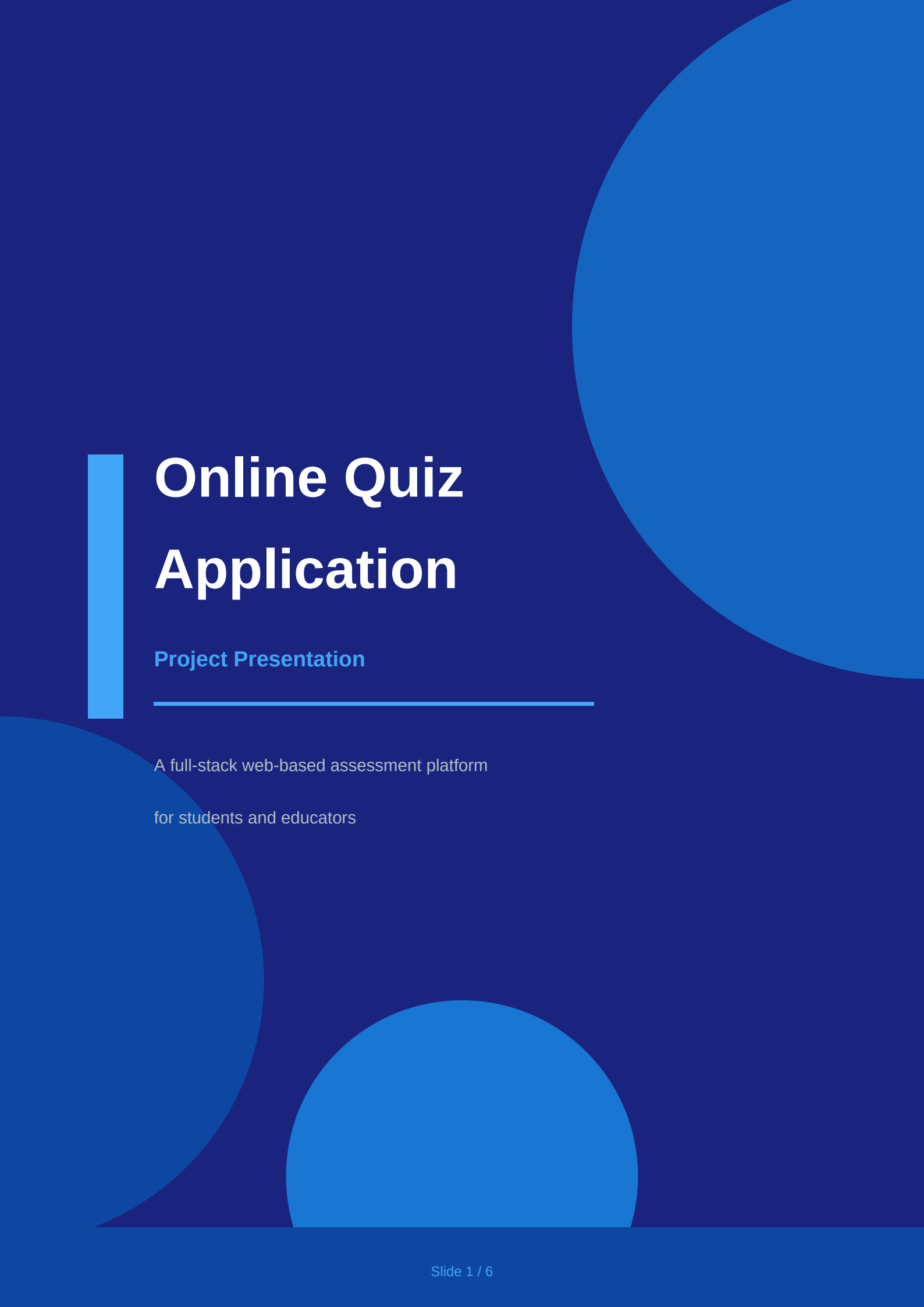




Online Quiz Application

Project Presentation

A full-stack web-based assessment platform

for students and educators

Project Overview

What is the Online Quiz Application?

The Online Quiz Application is a web-based platform that allows educators to create, manage, and publish quizzes, while students can attempt them in real time and receive instant feedback.

■ Purpose

- Automate assessments
- Save educator time
- Instant student feedback
- Track performance

■ Platform

- Web-based (browser)
- Responsive design
- Mobile friendly
- No installation needed

■ Target Users

- Students (all levels)
- Teachers / Educators
- Training institutes
- Corporate HR teams

Key Features

Core functionalities of the system

1

User Authentication

Secure login & registration for students and admins with role-based access control.

2

Quiz Management

Create, edit, publish quizzes with MCQ, True/False, and short-answer question types.

3

Timer & Auto-Submit

Countdown timer per quiz; auto-submits when time expires to ensure fairness.

4

Instant Result & Score

Scores calculated automatically; detailed result page shown immediately after submission.

5

Leaderboard

Real-time ranking board showing top performers for each quiz or category.

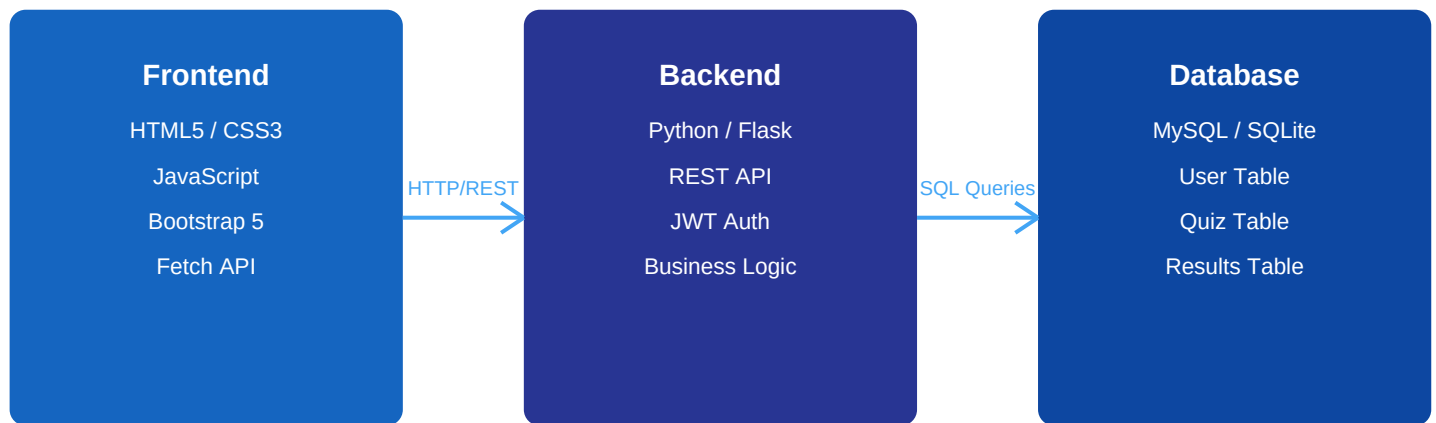
6

Admin Dashboard

Manage users, view attempt statistics, export reports as CSV.

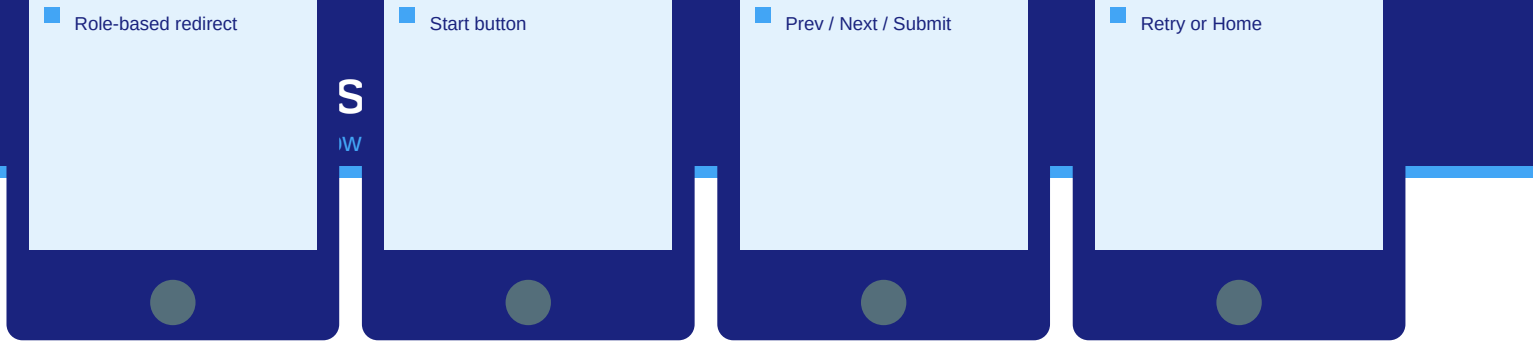
System Architecture

Technology stack & component design



Deployment: GitHub | Localhost / Render / PythonAnywhere | Environment: virtualenv + requirements.txt

Security: Password hashing (bcrypt) | JWT session tokens | Input validation & CSRF protection



Conclusion & Future Scope

Summary and upcoming enhancements

Project Summary

- Successfully built a fully functional web-based quiz platform.
- Implemented role-based access for Admin and Student users.
- Real-time scoring, timer, and leaderboard features delivered.
- Tested with multiple quiz categories and user sessions.

Future Enhancements

1. AI Question Generator

Auto-generate questions using LLM APIs based on topic input.

2. Mobile App (Flutter)

Cross-platform mobile version for Android and iOS devices.

3. Video Proctoring

Webcam-based monitoring to prevent cheating during exams.

4. Analytics Dashboard

Detailed performance graphs and heatmaps for educators.

5. Multi-language Support

Localize the platform in Tamil, Hindi, and other languages.

ER Diagram

Entity Relationship model of the database

